

during either stage of study selection were resolved through discussion and consensus, with consultation from a third reviewer when necessary. Studies were eligible if they evaluated educational and/or behavioral interventions designed to improve medication adherence among individuals with T2D and were published in English during the predefined study period. Medication adherence was required to be an explicit intervention target, outcome, or evaluation component. Broader diabetes self-management interventions were also eligible if medication adherence was assessed as either a primary or secondary outcome. Studies that focused exclusively on other self-management, such as diet, physical activity, foot care, or glycemic control, without evaluating medication adherence were excluded.

Data Charting and Synthesis

Data were charted using a predefined extraction form developed with reference to previous literature [33–35]. Extracted data included publication year, country, study design, participant characteristics, intervention components, and reported outcomes related to medication adherence or glycemic control. Because medication adherence was the primary focus of this review, information regarding adherence assessment methods was also extracted. Eligible studies measured adherence using a variety of approaches, including self-reported adherence questionnaires or scales, medication-taking records, refill-based measures, medication use logs, and intervention-specific adherence indicators. When reported, objective clinical outcomes associated with adherence, such as A1C or fasting plasma glucose, were also extracted. Data extraction was conducted by one reviewer and cross verified by other members of the research team to ensure accuracy. All extracted data were organized using Microsoft Excel.

A descriptive synthesis approach was undertaken to summarize study characteristics and intervention components. Frequencies and percentages were calculated to describe study design, geographic distribution, and intervention characteristics. To identify common behavioral approaches used to improve medication adherence, intervention descriptions were analyzed using inductive content analysis [36]. Intervention components were coded inductively according to their primary function and categorized into common behavioral strategies, including medication reminders, educational content, self-monitoring, patient-provider communication, peer support, telemonitoring, and personalized feedback.

Consistent with the objectives of scoping reviews, the purpose of this review was to map the breadth and characteristics of the available evidence rather than to evaluate intervention effectiveness or determine comparative efficacy. Therefore, a formal risk-of-bias or methodological

quality assessment was not conducted. This approach aligns with established scoping review methodology and should be considered when interpreting findings related to intervention effectiveness, as the included studies may vary in methodological rigor [32].

Results

Search Results

The initial search identified 2,236 records, of which 867 were duplicates. After screening titles and abstracts, 1,242 articles were excluded, leaving 127 studies for full text assessment. Of these, 71 were excluded due to lack of assessment of medication adherence, 11 were non-interventional studies, 9 did not involve individuals with type 2 diabetes, and 4 lacked sufficient detail on intervention components. Ultimately, 32 studies [4, 16, 18, 37, 39–42, 45–51, 53–56, 58–68] were included in the qualitative synthesis. Figure 1 illustrates the study selection process.

Study Characteristics

The included studies involved diverse populations and were conducted across multiple regions. Most studies were carried out in the United States ($n=10$, 31.3%) [4, 38, 41, 44–48, 52, 62], followed by Spain ($n=4$, 12.5%) [18, 39, 57, 65]. Additional studies were conducted in Asia ($n=14$, 43.8%) [40, 42, 46, 48–51, 53, 55, 58, 61, 63, 64], Africa ($n=2$, 6.3%) [16, 37], Australia ($n=2$, 6.3%) [49, 59], the Middle East ($n=2$, 6.3%) [56, 60], and the United Kingdom ($n=1$, 3.1%) [43]. Most studies used randomized controlled trial designs ($n=27$, 84.4%) [16, 18, 37, 38, 42–52, 54–65]. Four studies (12.5%) used a single-group pre–post-test design [39, 41, 53], and one study (3.1%) used a case-crossover design [4]. Sample sizes ranged from 13 to 1,119 participants. All studies included participants of both sexes, except for one study that enrolled only male veterans [4]. In addition, 43.8% of the studies reported a majority of male participants [4, 18, 43, 45, 48, 49, 51, 53, 55, 58–60, 64, 65] (Table 2).

Digital Intervention Components

The digital interventions were categorized according to both the digital platform used and the behavioral components incorporated within the intervention. The digital health tools identified in this review included mobile apps (delivered via smartphones, tablets, personal computers, or personal digital assistants), voice-based conversational artificial intelligence (AI), cloud-based platforms, remote